

Logical Reasoning for Computer Science

COMPSCI 2LC3

McMaster University, Fall 2026

Wolfram Kahl

2026-09-11

Expressions and Calculations

Plan for Today

- LADM 1.1: Syntax of Conventional Mathematical Expressions
- LADM 1.2: Substitution

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Part 1: Syntax of Mathematical Expressions

The Answer — Anything Wrong?

$$\begin{aligned}
 & 7 \cdot 8 \\
 = & \langle \text{Fact `8 = 7 + 1`} \rangle \\
 & 7 \cdot (7 + 1) \\
 = & \langle \text{Fact `7 = 10 - 3`} \rangle \\
 & (10 - 3) \cdot (7 + 1) \\
 = & \langle \text{"Distributivity of } \cdot \text{ over +"} \rangle \\
 & (10 - 3) \cdot 7 + (10 - 3) \cdot 1 \\
 = & \langle \text{"Distributivity of } \cdot \text{ over -"} \rangle \\
 & 10 \cdot 7 - 3 \cdot 7 + 10 \cdot 1 - 3 \cdot 1 \\
 = & \langle \text{"Identity of } \cdot \text{" — twice} \rangle \\
 & 10 \cdot 7 - 3 \cdot 7 + 10 - 3 \\
 = & \langle \text{Fact `3 \cdot 7 = 21`} \rangle \\
 & 10 \cdot 7 - 21 + 10 - 3 \\
 = & \langle \text{Fact `10 \cdot 7 = 70`} \rangle \\
 & 70 - 21 + 10 - 3 \\
 = & \langle \text{Fact `10 - 3 = 7`} \rangle \\
 & 70 - 21 + 7 \\
 = & \langle \text{Fact `21 + 7 = 28`} \rangle \\
 & 70 - 28 \\
 = & \langle \text{Fact `70 - 28 = 42`} \rangle \\
 & 42
 \end{aligned}$$

Syntax of Conventional Mathematical Expressions — For Now

LADM 1.1, p. 7

- A **constant** (e.g., 231) or **variable** (e.g., x) is an expression
- If E is an expression, then (E) is an expression
- If $\circ$ is a **unary prefix operator** and E is an expression, then $\circ E$ is an expression, with operand E .
For example, the negation symbol $-$ is used as a unary prefix operator, so -5 is an expression.
- If $\otimes$ is a **binary infix operator** and D and E are expressions, then $D \otimes E$ is an expression, with operands D and E .
For example, the symbols $+$ and $\cdot$ are binary infix operators, so $1 + 2$ and $(-5) \cdot (3 + x)$ are expressions.

("For now" — in particular no variable binders!)

Syntax of Conventional Mathematical Expressions

- A **constant** (e.g., 231) or **variable** (e.g., x) is an expression
- If E is an expression, then (E) is an expression
- If $\circ$ is a **unary prefix operator** and E is an expression, then $\circ E$ is an expression, with operand E .
- If $\otimes$ is a **binary infix operator** and D and E are expressions, then $D \otimes E$ is an expression, with operands D and E .

The intention of this is that each expression is **at least one** of the following alternatives:

- **either some constant**
- **or some variable**
- **or some simpler expression** in parentheses
- **or the application of some unary prefix operator**
to some simpler expression
- **or the application of some binary infix operator**
to two simpler expressions

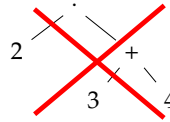
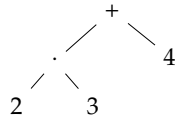
Why is this an expression?

$$2 \cdot 3 + 4$$

- If $\otimes$ is a **binary infix operator** and D and E are expressions, then $D \otimes E$ is an expression, with operands D and E .

- or the application of **some binary infix operator** to **two simpler expressions**

Which expression is it?



Why?

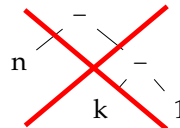
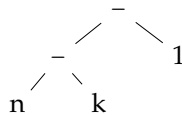
$\Rightarrow$ The multiplication operator $\cdot$ has **higher precedence** than the addition operator $+$.

Table of Precedences

- $[x := e]$ (textual substitution) **(highest precedence)**
- $\cdot$ (function application)
- unary prefix operators $+$, $-$, $\neg$, $\#$, $\sim$, $\mathcal{P}$
- $**$
- $\cdot$ $/$ $\div$ mod gcd
- $+$ $-$ $\cup$ $\cap$ $\times$ $\circ$ $\bullet$
- $\downarrow$ $\uparrow$
- $\#$
- $\triangleleft$ $\triangleright$ $\wedge$
- $=$ $<$ $>$ $\in$ $\subset$ $\subseteq$ $\supset$ $\supseteq$ $|$ (conjunctive)
- $\vee$ $\wedge$
- $\Rightarrow$ $\Leftarrow$
- $\equiv$ **(lowest precedence)**

All non-associative binary infix operators associate to the left, except $**$, $\triangleleft$, $\Rightarrow$, $\rightarrow$, which associate to the right.

Why is " $n - k - 1$ " an expressions? Which expression is it?



The operators $-$ **associates to the left**.

Associativity versus Association

- If we write $a + b + c$, there appears to be no need to discuss whether we mean $(a + b) + c$ or $a + (b + c)$, because they evaluate to the same values:

$$(a + b) + c = a + (b + c) \quad \text{"+" is associative}$$

- If we write $a - b - c$, we mean $(a - b) - c$:

$$\text{"-" associates to the left} \quad 9 - (5 - 2) \neq (9 - 5) - 2$$

- If we write a^{b^c} , we mean $a^{(b^c)}$:

$$\text{exponentiation associates to the right} \quad 2^{(3^2)} \neq (2^3)^2$$

- If we write $a ** b ** c$, we mean $a ** (b ** c)$:

$$\text{"**" associates to the right}$$

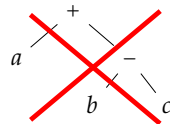
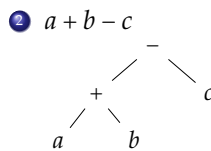
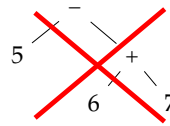
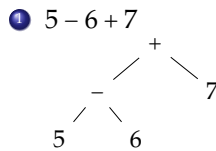
- If we write $a \Rightarrow b \Rightarrow c$, we mean $a \Rightarrow (b \Rightarrow c)$:

$$\text{"\Rightarrow" associates to the right} \quad (F \Rightarrow (T \Rightarrow F)) \neq ((F \Rightarrow T) \Rightarrow F)$$

- If we write $a \rightarrow b \rightarrow c$, we mean $a \rightarrow (b \rightarrow c)$:

$$\text{The function type constructor "\to" associates to the right} \quad (\mathbb{Z} \rightarrow (\mathbb{Z} \rightarrow \mathbb{B})) \neq ((\mathbb{Z} \rightarrow \mathbb{Z}) \rightarrow \mathbb{B})$$

Why are these expressions? Which expressions are these?



The operators + and - **associate to the left**, also mutually.

Precedences and Association — We write strings, but we think trees

All the rules we have for implicit parentheses only serve to encode the tree structure.

(We use underscores to denote operator argument positions.

So $_ \otimes _$ is a binary infix operator, and $_ \ominus _$ is a unary prefix operator.)

$_ \otimes _$ has higher precedence than $_ \ominus _$	means	$a \otimes b \ominus c = (a \otimes b) \ominus c$ $a \ominus b \otimes c = a \ominus (b \otimes c)$
$_ \otimes _$ has higher precedence than $_ \boxminus _$	means	$_ \boxminus a \otimes b = _ \boxminus (a \otimes b)$
$_ \boxminus _$ has higher precedence than $_ \otimes _$	means	$_ \boxminus a \otimes b = (_ \boxminus a) \otimes b$
$_ \otimes _$ associates to the left	means	$a \otimes b \otimes c = (a \otimes b) \otimes c$
$_ \otimes _$ associates to the right	means	$a \otimes b \otimes c = a \otimes (b \otimes c)$
$_ \otimes _$ mutually associates to the left with (same prec.) $_ \ominus _$	means	$a \otimes b \ominus c = (a \otimes b) \ominus c$
$_ \otimes _$ mutually associates to the right with (same prec.) $_ \ominus _$	means	$a \otimes b \ominus c = a \otimes (b \ominus c)$

Table of Precedences

- $[x := e]$ (textual substitution) (highest precedence)
- $.$ (function application)
- unary prefix operators $+$, $-$, $\neg$, $\#$, $\sim$, $\mathcal{P}$
- $**$
- $\cdot$ $/$ $\div$ $\bmod$ $\gcd$
- $+$ $-$ $\cup$ $\cap$ $\times$ $\circ$ $\bullet$
- $\downarrow$ $\uparrow$
- $\#$
- $\triangleleft$ $\triangleright$ $\wedge$
- $=$ $\neq$ $<$ $>$ $\in$ $\subset$ $\subseteq$ $\supset$ $\supseteq$ $|$ (conjunctive)
- $\vee$ $\wedge$
- $\Rightarrow$ $\nRightarrow$ $\Leftarrow$ $\nLeftarrow$
- $\equiv$ $\neq$ (lowest precedence)

All non-associative binary infix operators **associate to the left**,
except $**$, $\triangleleft$, $\Rightarrow$, $\rightarrow$, which **associate to the right**.

Conjunctive Operators

Chains can involve different conjunctive operators:

$$1 < i \leq j < 5 = k$$

$$\equiv \langle \text{"Reflexivity of "="} \setminus x = x \setminus \text{--- conjunctive operators} \rangle$$

$$1 < i \wedge i \leq j \wedge j < 5 \wedge 5 = k$$

$$\equiv \langle \text{"Reflexivity of "="} \text{---} \wedge \text{ has lower precedence} \rangle$$

$$(1 < i) \wedge (i \leq j) \wedge (j < 5) \wedge (5 = k)$$

$$x < 5 \in S \subseteq T$$

$$\equiv \langle \text{"Reflexivity of "="} \text{--- conjunctive operators} \rangle$$

$$x < 5 \wedge 5 \in S \wedge S \subseteq T$$

$$\equiv \langle \text{"Reflexivity of "="} \text{---} \wedge \text{ has lower precedence} \rangle$$

$$(x < 5) \wedge (5 \in S) \wedge (S \subseteq T)$$

Remember this!!!

Mathematical Expressions, Terms, Formulae ...

"Expression" is not the only word used for this kind of concept.

Related terminology:

- Both "term" and "expression" are frequently used names for the same kind of concept.
- The LADM's "expression" subsumes both "term" and "formula" of conventional first-order predicate logic.

Remember:

- Expressions are **understood** as tree-structures — "abstract syntax"
- Expressions are **written** as strings — "concrete syntax"
- Parentheses, precedences, and association rules **only serve to disambiguate the encoding of trees in strings.**

An Equational Theory of Integers — Axioms (LADM Ch. 15)

- (15.1) **Axiom, Associativity:** $(a + b) + c = a + (b + c)$
 $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- (15.2) **Axiom, Symmetry:** $a + b = b + a$
 $a \cdot b = b \cdot a$
- (15.3) **Axiom, Additive identity:** $0 + a = a$
 $a + 0 = a$
- (15.4) **Axiom, Multiplicative identity:** $1 \cdot a = a$
 $a \cdot 1 = a$
- (15.5) **Axiom, Distributivity:** $a \cdot (b + c) = a \cdot b + a \cdot c$
 $(b + c) \cdot a = b \cdot a + c \cdot a$
- (15.13) **Axiom, Unary minus:** $a + (-a) = 0$
- (15.14) **Axiom, Subtraction:** $a - b = a + (-b)$

An Equational Theory of Integers — Axioms (CALC CHECK)

Ex1.2 Starting Point

Declaration: $\mathbb{Z}$: Type

Declaration: $_ + _ : \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}$

Declaration: $_ \cdot _ : \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}$

Axiom (15.1) (15.1a) "Associativity of +": $(a + b) + c = a + (b + c)$

Axiom (15.1) (15.1b) "Associativity of ·": $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

Axiom (15.2) (15.2a) "Symmetry of +": $a + b = b + a$

Axiom (15.2) (15.2b) "Symmetry of ·": $a \cdot b = b \cdot a$

Axiom (15.3) "Additive identity" "Identity of +": $0 + a = a$

Axiom (15.4) "Multiplicative identity" "Identity of ·": $1 \cdot a = a$

Axiom (15.5) "Distributivity of · over +": $a \cdot (b + c) = a \cdot b + a \cdot c$

Axiom (15.9) "Zero of ·": $a \cdot 0 = 0$

Declaration: $_ - _ : \mathbb{Z} \rightarrow \mathbb{Z}$

Declaration: $_ - _ : \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}$

Axiom (15.13) "Unary minus": $a + (-a) = 0$

Axiom (15.14) "Subtraction": $a - b = a + (-b)$

Calculational Proofs of Theorems — (15.17) $-(-a) = a$

(15.3) Identity of + $0 + a = a$	(15.13) Unary minus $a + (-a) = 0$
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LADM:

Theorem (15.17): $-(-a) = a$

Proof:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{Identity of + (15.3)} \rangle \\
 & 0 + -(-a) \\
 = & \langle \text{Unary minus (15.13)} \rangle \\
 & a + (-a) + -(-a) \\
 = & \langle \text{Unary minus (15.13)} \rangle \\
 & a + 0 \\
 = & \langle \text{Identity of + (15.3)} \rangle \\
 & a
 \end{aligned}$$

CALC CHECK:

Theorem (15.17) "Self-inverse of unary minus":

$$-(-a) = a$$

Proof:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{"Identity of +"} \rangle \\
 & 0 + -(-a) \\
 = & \langle \text{"Unary minus"} \rangle \\
 & a + (-a) + -(-a) \\
 = & \langle \text{"Unary minus"} \rangle \\
 & a + 0 \\
 = & \langle \text{"Identity of +"} \rangle \\
 & a
 \end{aligned}$$

in Ex1.2

(15.17) with Explicit Associativity and Symmetry Steps

(15.3) Identity of + $0 + a = a$	(15.13) Unary minus $a + (-a) = 0$
----------------------------------	------------------------------------

Proving (15.17) $-(-a) = a$:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{Identity of + (15.3)} \rangle \\
 & 0 + -(-a) \\
 = & \langle \text{Unary minus (15.13)} \rangle \\
 & (a + (-a)) + -(-a) \\
 = & \langle \text{Associativity of + (15.1)} \rangle \\
 & a + ((-a) + -(-a)) \\
 = & \langle \text{Unary minus (15.13)} \rangle \\
 & a + 0 \\
 = & \langle \text{Symmetry of + (15.2)} \rangle \\
 & 0 + a \\
 = & \langle \text{Identity of + (15.3)} \rangle \\
 & a
 \end{aligned}$$

Automatic Application of Associativity and Symmetry Laws

Axiom (15.1) (15.1a) "Associativity of +": $(a + b) + c = a + (b + c)$

Axiom (15.1) (15.1b) "Associativity of ·": $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

Axiom (15.2) (15.2a) "Symmetry of +": $a + b = b + a$

Axiom (15.2) (15.2b) "Symmetry of ·": $a \cdot b = b \cdot a$

- You have been trained to reason "up to symmetry and associativity"
- Making symmetry and associativity steps explicit is
 - **always allowed**
 - sometimes **very useful for readability**
- CALCCHECK allows selective activation of symmetry and associativity laws
 - ⇒ "Exercise ... / Assignment ...: [...] **without automatic associativity and symmetry**"
 - ⇒ Having to make symmetry and associativity steps explicit can be tedious...

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Part 2: Substitution

Textual Substitution

Let E and R be expressions **without variable binders** and let x be a variable. We write:

$$E[x := R]$$

to denote an expression that is the same as E but with all occurrences of x replaced by (R) . (LADM very occasionally writes E_R^x instead of $E[x := R]$.)

Example 1:

$$\begin{aligned} & (x + y)[x := z + 2] \\ = & \langle \text{Substitution — performing substitution} \rangle \\ & ((z + 2) + y) \\ = & \langle \text{“Reflexivity of =” — removing unnecessary parentheses} \rangle \\ & z + 2 + y \end{aligned}$$

Textual Substitution

Let E and R be expressions and let x be a variable. We write:

$$E[x := R]$$

to denote an expression that is the same as E but with all occurrences of x replaced by (R) .

Example 2:

$$\begin{aligned} & (x \cdot y)[x := z + 2] \\ = & \langle \text{Substitution} \rangle \\ & ((z + 2) \cdot y) \\ = & \langle \text{“Reflexivity of =” — removing unnecessary parentheses} \rangle \\ & (z + 2) \cdot y \end{aligned}$$

Parentheses are needed here to keep the subtree corresponding to R together!

Textual Substitution

Let E and R be expressions and let x be a variable. We write:

$$E[x := R]$$

to denote an expression that is the same as E but with all occurrences of x replaced by (R) .

Example 3:

$$\begin{aligned} & (0 + a)[a := -(-a)] \\ = & \langle \text{Substitution} \rangle \\ & (0 + (-(-a))) \\ = & \langle \text{“Reflexivity of =” — removing (some) unnecessary parenth.} \rangle \\ & 0 + -(-a) \end{aligned}$$

Textual Substitution

Let E and R be expressions and let x be a variable. We write:

$$E[x := R]$$

to denote an expression that is the same as E but with all occurrences of x replaced by (R) .

Example 4:

$$\begin{aligned} & x + y[x := z + 2] \\ = & \langle \text{"Reflexivity of =" — adding parentheses for clarity} \rangle \\ & x + (y[x := z + 2]) \\ = & \langle \text{Substitution} \rangle \\ & x + (y) \\ = & \langle \text{"Reflexivity of =" — removing unnecessary parentheses} \rangle \\ & x + y \end{aligned}$$

Note: Substitution $[x := R]$ is a **highest precedence** postfix operator

Textual Substitution

Let E and R be expressions and let x be a variable. We write:

$$E[x := R] \quad \text{or} \quad E_R^x$$

to denote an expression that is the same as E but with all occurrences of x replaced by (R) .

Examples:

Expression	Result	Unnecessary parentheses removed
$x[x := z + 2]$	$(z + 2)$	$z + 2$
$(x + y)[x := z + 2]$	$((z + 2) + y)$	$z + 2 + y$
$(x \cdot y)[x := z + 2]$	$((z + 2) \cdot y)$	$(z + 2) \cdot y$
$x + y[x := z + 2]$	$x + y$	$x + y$

Note: Substitution $[x := R]$ is a **highest precedence** postfix operator

Sequential Substitution

$$\begin{aligned} & (x + y)[x := y - 3][y := z + 2] \\ = & \langle \text{"Reflexivity of =" — adding parentheses for clarity} \rangle \\ & ((x + y)[x := y - 3])[y := z + 2] \\ = & \langle \text{Substitution — performing inner substitution} \rangle \\ & (((y - 3) + y))[y := z + 2] \\ = & \langle \text{Substitution — performing outer substitution} \rangle \\ & (((z + 2) - 3) + (z + 2)) \\ = & \langle \text{"Reflexivity of =" — removing unnecessary parentheses} \rangle \\ & z + 2 - 3 + z + 2 \end{aligned}$$

(The technical reason for performing the inner substitution first is that substitution itself is a variable binder, and the recipe for performing substitutions $E[x := R]$ given in LADM chapter 1 cannot be used once E contains variable binders.)

Simultaneous Textual Substitution

If R is a **list** $R_1, \dots, R_n$ of expressions
and x is a **list** $x_1, \dots, x_n$ of **distinct variables**, we write:

$$E[x := R]$$

to denote the **simultaneous** replacement of the variables of x
by the corresponding expressions of R ,
each expression being enclosed in parentheses.

Example:

$$\begin{aligned} & (x + y)[x, y := y - 3, z + 2] \\ &= \langle \text{Substitution — performing substitution} \rangle \\ & \quad ((y - 3) + (z + 2)) \\ &= \langle \text{“Reflexivity of =” — removing unnecessary parentheses} \rangle \\ & \quad y - 3 + z + 2 \end{aligned}$$

Simultaneous Textual Substitution

If R is a **list** $R_1, \dots, R_n$ of expressions
and x is a **list** $x_1, \dots, x_n$ of **distinct variables**, we write:

$$E[x := R]$$

to denote the **simultaneous** replacement of the variables of x
by the corresponding expressions of R ,
each expression being enclosed in parentheses.

Examples:

Expression	Result	Unnecessary parentheses removed
$x[x, y := y - 3, z + 2]$	$(y - 3)$	$y - 3$
$(y + x)[x, y := y - 3, z + 2]$	$((z + 2) + (y - 3))$	$z + 2 + y - 3$
$(x + y)[x, y := y - 3, z + 2]$	$((y - 3) + (z + 2))$	$y - 3 + z + 2$
$x + y[x, y := y - 3, z + 2]$	$x + (z + 2)$	$x + z + 2$

Simultaneous Substitution:

$$\begin{aligned} & (x + y)[x, y := y - 3, z + 2] \\ &= \langle \text{Substitution — performing substitution} \rangle \\ & \quad ((y - 3) + (z + 2)) \\ &= \langle \text{“Reflexivity of =” — removing unnecessary parentheses} \rangle \\ & \quad y - 3 + z + 2 \end{aligned}$$

Sequential Substitution:

$$\begin{aligned} & (x + y)[x := y - 3][y := z + 2] \\ &= \langle \text{“Reflexivity of =” — adding parentheses for clarity} \rangle \\ & \quad ((x + y)[x := y - 3])[y := z + 2] \\ &= \langle \text{Substitution — performing inner substitution} \rangle \\ & \quad (((y - 3) + y))[y := z + 2] \\ &= \langle \text{Substitution — performing outer substitution} \rangle \\ & \quad (((z + 2) - 3) + (z + 2)) \\ &= \langle \text{“Reflexivity of =” — removing unnecessary parentheses} \rangle \\ & \quad z + 2 - 3 + z + 2 \end{aligned}$$

Calculational Proofs of Theorems — (15.17) — $-(-a) = a$

(15.3) Identity of + $0 + a = a$	(15.13) Unary minus $a + (-a) = 0$
----------------------------------	------------------------------------

Theorem (15.17) “Self-inverse of unary minus”: $-(-a) = a$

Proof:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{Identity of + (15.3)} \rangle \\
 & 0 + -(-a) \\
 = & \langle \text{Unary minus (15.13)} \rangle \\
 & a + (-a) + -(-a) \\
 = & \langle \text{Unary minus (15.13)} \rangle \\
 & a + 0 \\
 = & \langle \text{Identity of + (15.3)} \rangle \\
 & a
 \end{aligned}$$

Three different variables named “a”!

Calculational Proofs of Theorems — (15.17) — Renamed Theorem Variables

(15.3x) Identity of + $0 + x = x$	(15.13y) Unary minus $y + (-y) = 0$
-----------------------------------	-------------------------------------

Theorem (15.17) “Self-inverse of unary minus”: $-(-a) = a$

Proof:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{Identity of + (15.3x)} \rangle \\
 & 0 + -(-a) \\
 = & \langle \text{Unary minus (15.13y)} \rangle \\
 & a + (-a) + -(-a) \\
 = & \langle \text{Unary minus (15.13y)} \rangle \\
 & a + 0 \\
 = & \langle \text{Identity of + (15.3x)} \rangle \\
 & a
 \end{aligned}$$

Three different variables “x”, “y”, “a”.

Details of Applying Theorems — (15.17) with Explicit Substitutions I

(15.3x) Identity of + $0 + x = x$	(15.13y) Unary minus $y + (-y) = 0$
-----------------------------------	-------------------------------------

Theorem (15.17) “Self-inverse of unary minus”: $-(-a) = a$

Proof:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{Identity of + (15.3x) with } x := -(-a) \rangle \quad \boxed{(0 + x = x)[x := -(-a)] = (0 + -(-a) = -(-a))} \\
 & 0 + -(-a) \\
 = & \langle \text{Unary minus (15.13y) with } y := a \rangle \quad \boxed{(y + (-y) = 0)[y := a] = (a + (-a) = 0)} \\
 & a + (-a) + -(-a) \\
 = & \langle \text{Unary minus (15.13y) with } y := -a \rangle \quad \boxed{(y + (-y) = 0)[y := -a] = (-a + (-(-a)) = 0)} \\
 & a + 0 \\
 = & \langle \text{Identity of + (15.3x) with } x := a \rangle \quad \boxed{(0 + x = x)[x := a] = (0 + a = a)} \\
 & a
 \end{aligned}$$

Details of Applying Theorems — (15.17) with Explicit Substitutions II

(15.3) Identity of + $0 + a = a$ (15.13) Unary minus $a + (-a) = 0$

Theorem (15.17) “Self-inverse of unary minus”: $-(-a) = a$

Proof:

$$\begin{aligned}
 & -(-a) \\
 = & \langle \text{Identity of + (15.3) with } a := -(-a) \rangle \\
 & 0 + -(-a) \\
 = & \langle \text{Unary minus (15.13) with } a := a \rangle \\
 & a + (-a) + -(-a) \\
 = & \langle \text{Unary minus (15.13) with } a := -a \rangle \\
 & a + 0 \\
 = & \langle \text{Identity of + (15.3) with } a := a \rangle \\
 & a
 \end{aligned}$$

Three different variables named “a”!

Specifying Substitutions for Theorem Application in CalcCHECK

Theorem (15.19) “Distributivity of unary minus over +”: $-(a + b) = (-a) + (-b)$

Proof:

$$\begin{aligned}
 & -(a + b) \\
 = & \langle (15.20) \text{ with } a := a + b \rangle \\
 & (-1) \cdot (a + b) \\
 = & \langle \text{“Distributivity of } \cdot \text{ over +” with } a, b, c := -1, a, b \rangle \\
 & (-1) \cdot a + (-1) \cdot b \\
 = & \langle (15.20) \text{ with } a := b \rangle \\
 & (-1) \cdot a + -b \\
 = & \langle (15.20) \text{ with } a := a \rangle \\
 & (-a) + (-b)
 \end{aligned}$$

Theorem (15.20):

$$-a = (-1) \cdot a$$

- Backquotes enclose math embedded in English. (Markdown convention)
- Substitution notation as in LADM: $variables := expressions$
- “:=” reads “becomes” or “is/are replaced with”
- “:=” is entered by typing “\:=” or “\becomes”!
- The variable list has the same length as the expression list.
- No variable occurs twice in the variable list.
- CalcCHECK_{Web} notebooks “with rigid matching” **require** all theorem variables to be substituted. “Rigid matching” means: The theorems you specify need to match without substitution.

Coming Up: Foundations of Applying Equations in Context

Reasoning about Assignment Commands in Imperative Programs ($\approx$ LADM 1.6)

Anatomy of calculation based on Substitution (LADM 1.3–1.5):

- **Inference rule Substitution:** Justifies applying instances of theorems:

$$\begin{aligned}
 & 2 \cdot y + -(2 \cdot y) \\
 = & \langle \text{“Unary minus” } a + -a = 0 \text{ with } a := 2 \cdot y \rangle \\
 & 0
 \end{aligned}$$

- **Inference rule Leibniz:** Justifies applying (instances of) **equational** theorems deeper inside expressions:

$$\begin{aligned}
 & 2 \cdot x + 3 \cdot (y - 5 \cdot (4 \cdot x + 7)) \\
 = & \langle \text{“Subtraction” } a - b = a + -b \text{ with } a, b := y, 5 \cdot (4 \cdot x + 7) \rangle \\
 & 2 \cdot x + 3 \cdot (y + -(5 \cdot (4 \cdot x + 7)))
 \end{aligned}$$

H2 Starting Point

$$\begin{aligned}
 & 7 \cdot 8 \\
 = & \langle \text{Fact `8 = 7 + 1`} \rangle \\
 & 7 \cdot (7 + 1) \\
 = & \langle \text{Fact `7 = 10 - 3`} \rangle \\
 & (10 - 3) \cdot (7 + 1) \\
 = & \langle \text{"Distributivity of } \cdot \text{ over +"} \rangle \\
 & (10 - 3) \cdot 7 + (10 - 3) \cdot 1 \\
 = & \langle \text{"Distributivity of } \cdot \text{ over -"} \rangle \\
 & 10 \cdot 7 - 3 \cdot 7 + 10 \cdot 1 - 3 \cdot 1 \\
 = & \langle \text{"Identity of } \cdot \text{" — twice} \rangle \\
 & 10 \cdot 7 - 3 \cdot 7 + 10 - 3 \\
 = & \langle \text{Fact `3 \cdot 7 = 21`} \rangle \\
 & 10 \cdot 7 - 21 + 10 - 3 \\
 = & \langle \text{Fact `10 \cdot 7 = 70`} \rangle \\
 & 70 - 21 + 10 - 3 \\
 = & \langle \text{Fact `10 - 3 = 7`} \rangle \\
 & 70 - 21 + 7 \\
 = & \langle \text{Fact `21 + 7 = 28`} \rangle \\
 & 70 - 28 \\
 = & \langle \text{Fact `70 - 28 = 42`} \rangle \\
 & 42
 \end{aligned}$$

Next Steps

- Submit H2 by 12:30 on Monday, Sept. 14
- Work through Homework 2 before your tutorial, if that is earlier!
- Get started working on Exercises 1.*
- Go to your tutorial to continue working on Ex1 — bring your laptop!